Ziliang Xiong, Ph.D. Student

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Research Scientist
PhD Intern – 2027
Multimodal Computer Vision

Research Profile

- Final-year Ph.D. candidate in computer vision at Linköping University, specializing in **uncertainty quantification** for deep learning. Research spans regression, human pose estimation, autonomous driving, and multimodal vision-language-action models.
- Published at **CVPR Workshops**, **Pattern Recognition Letters**, and **ICASSP**; experienced in research design, Python/PyTorch prototyping, multi-GPU evaluation, and scientific writing.
- Available **full-time for 13–17 weeks from May 2027**; start date flexible.

Education

Sep 2022 – Aug 2027 **Ph.D. in Computer Vision, Linköping University, Sweden**
Based at *Computer Vision and Learning Systems*, Department of Electrical Engineering; fully funded by ELLIIT[1] and affiliated with the WASP[2] Graduate School.

Oct 2026 – Mar 2027 **Visiting Researcher, Delft University of Technology, the Netherlands**
Researching multimodal calibration for vision-language-action models in autonomous driving with Prof. Holger Caesar.

Sep 2020 – Jun 2022 **M.Sc. in Machine Learning, Systems and Control, Lund University, Sweden**
Master's thesis on deep-learning-based millimeter-wave radar detection for autonomous driving; scholarship awarded by Zenseact AB.

Sep 2015 – Jun 2019 **B.Sc. in Automation, Beihang University, China**, Shenyuan Honors Degree

Research & Project Highlights

2026 – Now **Multimodal Calibration and Risk Assessment for Vision-Language-Action Models**
- Developing calibration and risk-assessment methods for autonomous-driving VLA models with TU Delft, supported by 380K SEK in Vinnova FFI and WASP funding.

2023 – 2024 **Uncertainty Quantification Metrics for Deep Regression**
- Co-first-authored the study with equal contribution; cited 59 times.
- Aimed to identify robust uncertainty-quantification metrics for regression tasks by benchmarking 4 metrics under 4 uncertainty types.
- Revealed the distinction between calibration and ranking tasks in uncertainty evaluation.

2024 – 2025 **Collision Risk Estimation for End-to-End Autonomous Driving**
- First-authored and led a 6-researcher study; under major revision at IJCV.
- Aimed to assess whether end-to-end driving models' internal representations encode the risks of their own decisions.
- RiskMonitor improved collision avoidance by 66.5% in closed-loop tests on safety-critical scenarios.

2021 **Perspective Auto-Calibration, Axis Communications AB**
- Built perspective-regression modules to post-process object detections and reduce false positives in security-camera applications.
- Created an internal video dataset and a visualization tool for quantitative and qualitative model evaluation.

Vision-Based Weed Detection for FarmBot, Lund University Robotics Lab
- Trained YOLOv3 on a custom dataset of 250+ labeled plant images across 10 classes, achieving 83.2% test accuracy.
- Built the perception pipeline from plant scanning and weed detection to projection into robot coordinates for removal.

[1] A Swedish government-funded strategic research environment in information technology and mobile communications.
[2] The Wallenberg AI, Autonomous Systems and Software Program, Sweden's major national AI research initiative.

Research Publications

1. **Z. Xiong**, H. Shi, and P.-E. Forssen, "Meom: Multi-view expected-oks maximization for human pose triangulation," *arXiv preprint arXiv:2608.30521*, 2026, Under review at WACV 2027.
2. **Z. Xiong**, S. Wang, and H. Caesar, "Multimodal calibration for vision-language-action models in autonomous driving," Manuscript under preparation, 2026.
3. T. Dehdarirad, M. Felsberg, G. Eilertsen, and **Z. Xiong**, "Commutator-induced uncertainty in VAEs," Under review at ICLR 2027, 2026.
4. S. Liu, **Z. Xiong**, K.-H. Ngo, and P.-E. Forssén, "Matter: Multiscale attention for registration error regression," *2026 IEEE International Conference on Acoustics, Speech, and Signal Processing*, 2026.
5. **Z. Xiong**, S. Liu, N. Helgesen, H. Li, J. Johnander, and P.-E. Forssen, "Collision risk estimation via loss prediction in end-to-end autonomous driving," *arXiv preprint arXiv:2503.07425*, 2025, Major revision at International Journal of Computer Vision (IJCV).
6. S. Liu, **Z. Xiong**, B. Wandt, and P.-E. Forssén, "Continuous flow modeling for uncertainty-aware human pose estimation," *Scandinavian Conference on Image Analysis*, 2025.
7. **Z. Xiong**, A. Jonnarth, A. Eldesokey, J. Johnander, B. Wandt, and P.-E. Forssén, "Hinge-wasserstein: Estimating multimodal aleatoric uncertainty in regression tasks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, Jun. 2024, pp. 3471–3480.
8. **Z. Xiong**, S. Kristoffersson Lind, P.-E. Forssén, and V. Krüger, "Uncertainty quantification metrics for deep regression," *Pattern Recognition Letters*, vol. 186, pp. 91–97, 2024, ISSN: 0167-8655. DOI: https://doi.org/10.1016/j.patrec.2024.09.011.

Funding & Honors

2026 — ***Vinnova FFI Project Funding***, co-applicant and main researcher, VLA risk assessment for automated vehicles, 240K SEK

2026 – 2027 — **WASP Research Stint Abroad Grant**, TU Delft, 140K SEK

2022 – 2027 — **Fully Funded Ph.D. Position**, ELLIIT Project C08, safe autonomous robotic systems

Honors — **Long Oral Certificate**, CVPR SAIAD Workshop (2024); **Scholarship for Excellent Study**, Beihang University (2016–2019)

Employment

Sep 2022 – Now — **Master's Thesis Supervisor, Teaching Assistant, and Laboratory Assistant, Linköping University**
Supervise master's theses and serve as TA in Machine Learning for Computer Vision, Embedded Perception Systems, Signal and System, Images and Graphics, and Project Course.

Sep 2021 – Nov 2021 — **SI-PASS Leader, European Centre for SI-PASS, Lund University**
Led supplementary instruction sessions for the Image Analysis course.

Mar 2021 – Nov 2021 — **Student Project Developer, Lund University Robotics Lab** (see FarmBot project highlight above)

Jun 2021 – Sep 2021 — **Machine Learning Summer Intern, Axis Communications AB** (see project highlight above)

Technical Skills

Research — Computer vision, deep learning, uncertainty quantification, multimodal learning, autonomous driving, experimental design

Programming — **Python, PyTorch**, TensorFlow, MATLAB, Git, SQL, Julia, Apptainer, LaTeX

Languages — English (professional), Mandarin Chinese (native), Swedish (A1)